Modelling Project Report:

Accumulator (Battery) Storage Technologies for Renewable Energy Systems

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1. **Milestone 1:**

   In Milestone 1, we researched about the battery and storage technologies. The research included the following battery types:
   - Lead/Acid battery
   - Lithium-Ion (Li-ion) battery
   - Salt-Water battery (Aqueous or NaCl-based)
   - Redox Flow battery (RFB)

   We researched the specifications of each battery, and individual real-life cases involving each battery was also presented. A table was also

derived comparing the technologies and features of each battery, and emerging trends in battery/electrical storage technologies were also discussed, which included:

- Sodium-Ion batteries becoming industrially viable

- Solid-state batteries with solid electrolytes
- Alternative/Hybrid chemistries

2. Milestone 2:

In Milestone 2, a literature review was conducted on accumulator/battery models and their relevant trends over the past 5 years, in addition to analysis being done on several MATLAB models in order to select the most feasible model.

A. Literature Review:

The literature review conducted would lead to the discovery of the following models:

- Equivalent-Circuit Models (ECMs) → Representing the battery as a combination of Open Circuit Voltage (OCV) and one or more resistor/resistor-capacitor (RC) networks.

- Behavioural and table-based model → Rely on tabular relationship between SOC and OCV, may

include empirical functions/lookup tables for internal resistance, self-discharge and charge dynamics. Such examples are the Battery (Table-Based) and Generic Battery blocks in Simscape Electrical, which define discharge curves, self-discharge resistance, and temperature effects.
- Data-driven and hybrid models → Combines neural networks with ECMs, these are powerful for estimating state of charge (SOC), state of health (SOH), and remaining useful life (RUL), but these require large datasets and are less practical for Simulink/Simscape analysis.
- High-fidelity Electrochemical Models → describe ion transport and reaction kinetics through coupled equations. Highly accurate, but require computational power, making the model unfeasible for large-scale PV/Wind system analysis.

Recent studies published between 2020 and 2025 highlight a clear trend towards hybrid and data-driven modelling approaches, combined with advanced parameter identification techniques. For instance, a 2023

review of SOC/SOH estimation for lithium-ion batteries emphasizes the growing use of neural-network and machine-learning methods blended with simplified equivalent-circuit models [1]. Similarly, a 2024 investigation on equivalent-circuit modelling for battery degradation and ageing presents integrated frameworks linking thermal and ageing phenomena with circuit parameters. Recent works also employ electrochemical impedance spectroscopy (EIS) and distribution-of-relaxation-time (DRT) analyses to derive improved ECMs that capture charge/discharge dynamics over a wide range of SOC and temperature conditions [3]. These developments confirm that, within the past five years, the modelling community has shifted from purely empirical representations to hybrid and physics-informed methods capable of accurately describing both dynamic (charging/discharging) and long-term (thermal/aging) battery behaviours, while maintaining computational feasibility in MATLAB/Simulink environments.

B. Analysis of several MATLAB models:

The following MATLAB/Simulink models were analyzed for selection:

- **Empirical Rint/Shepherd Model→ Implemented with standard Simulink blocks (Integrator, Lookup Table, Sum, Gain). Requires few parameters and is ideal for quick feasibility studies or as a benchmark.**
- **Thevenin Equivalent-Circuit (EC) Model → Extends the Rint model by adding an RC branch for dynamic voltage response. Can be created with a Transfer Function block or implemented through the Simscape Battery**

  "Battery Equivalent Circuit" component.

  Provides a balance between computational simplicity and physical realism.
- **Simscape Battery ECM with Thermal/Aging → Adds temperature-dependent parameters, self-discharge resistance, and lifetime effects. Useful for long-term techno-economic studies of PV/wind systems but requires detailed manufacturer data.**

Model selected → Thevenin EC Model.

3. Milestone 3:

In Milestone 3, the model would be implemented on MATLAB/Simulink. In this part, we will be talking about the equations of the model in addition to how these equations were implemented on Simulink.

A. Model Equations:

The equations derived from the model are as follows:

$$\frac{dSOC}{dt} = -\frac{I(t)}{Q_{\text{nom}}} \tag{1}$$

$$\frac{dV_p}{dt} = -\frac{1}{R_1 C_1} V_p + \frac{1}{C_1} I(t) \tag{2}$$

$$V_t = E(SOC) - I(t) R_0 - V_p \tag{3}$$

where:

- $E(SOC)$: Open-circuit voltage as a function of SOC
- R_0: Ohmic resistance
- R_1, C_1: RC branch parameters
- V_p: Polarization voltage

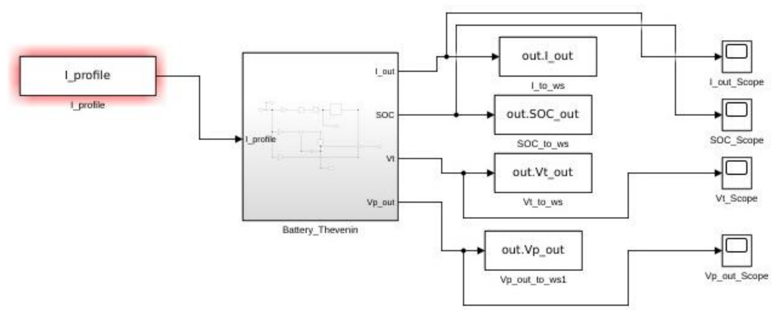
Parameters required:

- Nominal capacity Q_{nom} [Ah]
- OCV–SOC curve (from experimental or literature data)
- Resistances R_0, R_1 and capacitance C_1
- SOC and temperature operating limits

B. Simulink Implementation:

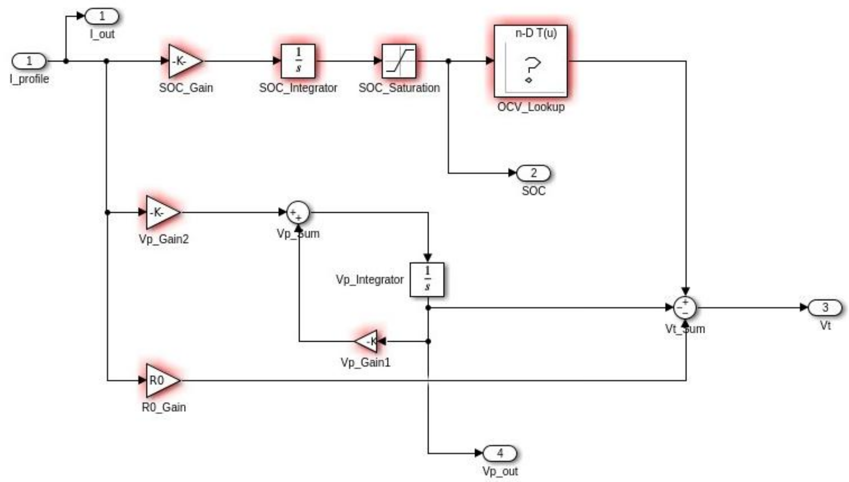
The model was implemented on Simulink with the following steps:

1. **Defining parameters, inputs, Simulink sub-circuit, and outputs → Input defined was the I_profile as a series of values changing over time (a lower value till 60s then between 60s and 120s the I_profile switches to a higher value)**
2. **Implementation of Simulink subcircuit which converts the input into 4 outputs: I_out, SOC, Vt, and Vp_out. Please take a look at the picture below for further clarification:**



The sub-circuit was implemented as follows (Please kindly ignore the red errors since that was caused by the I_profile being deleted from the MATLAB workspace, however, when you

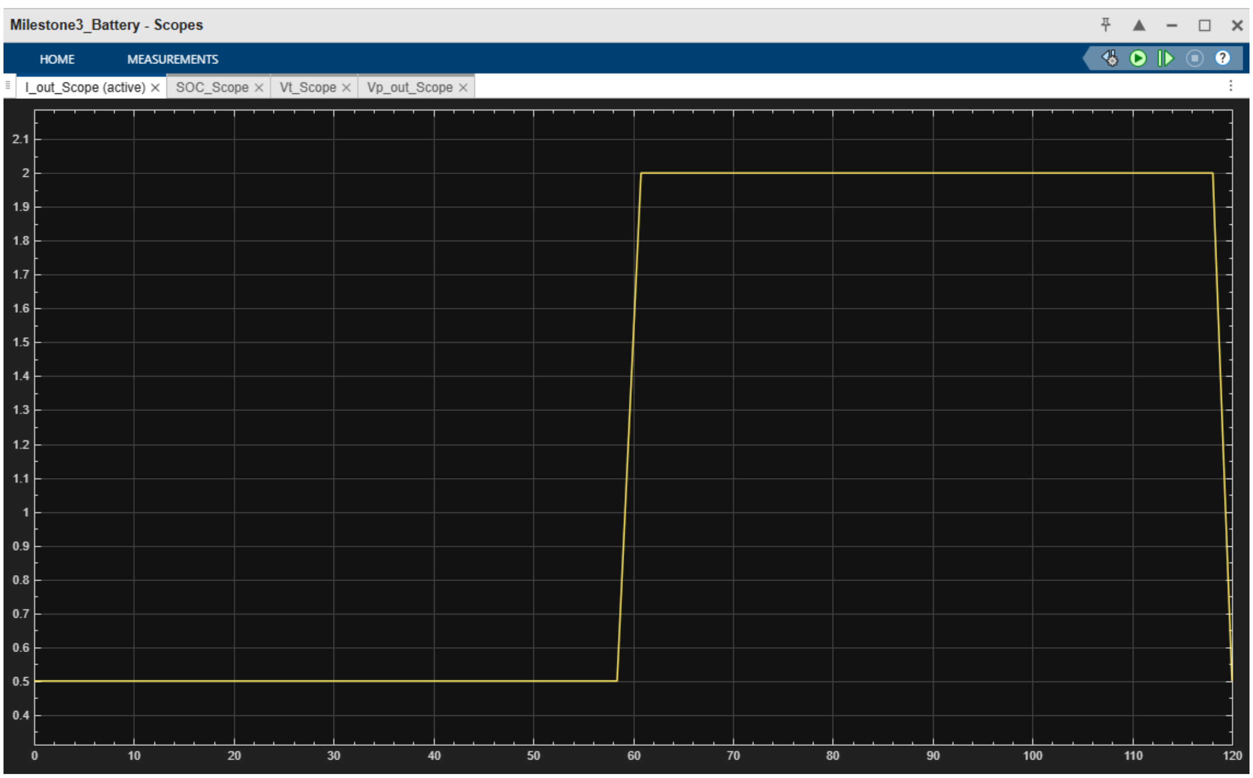
run all the attached codes, the I_profile is defined):

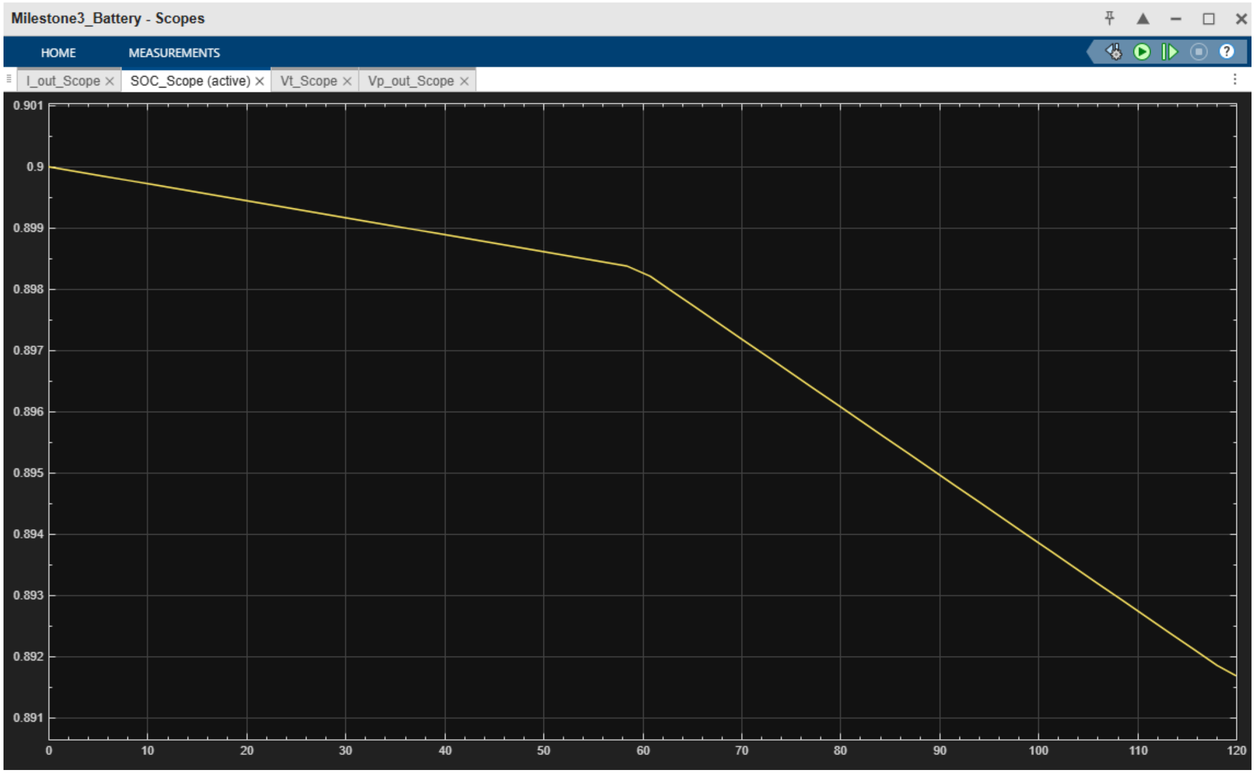


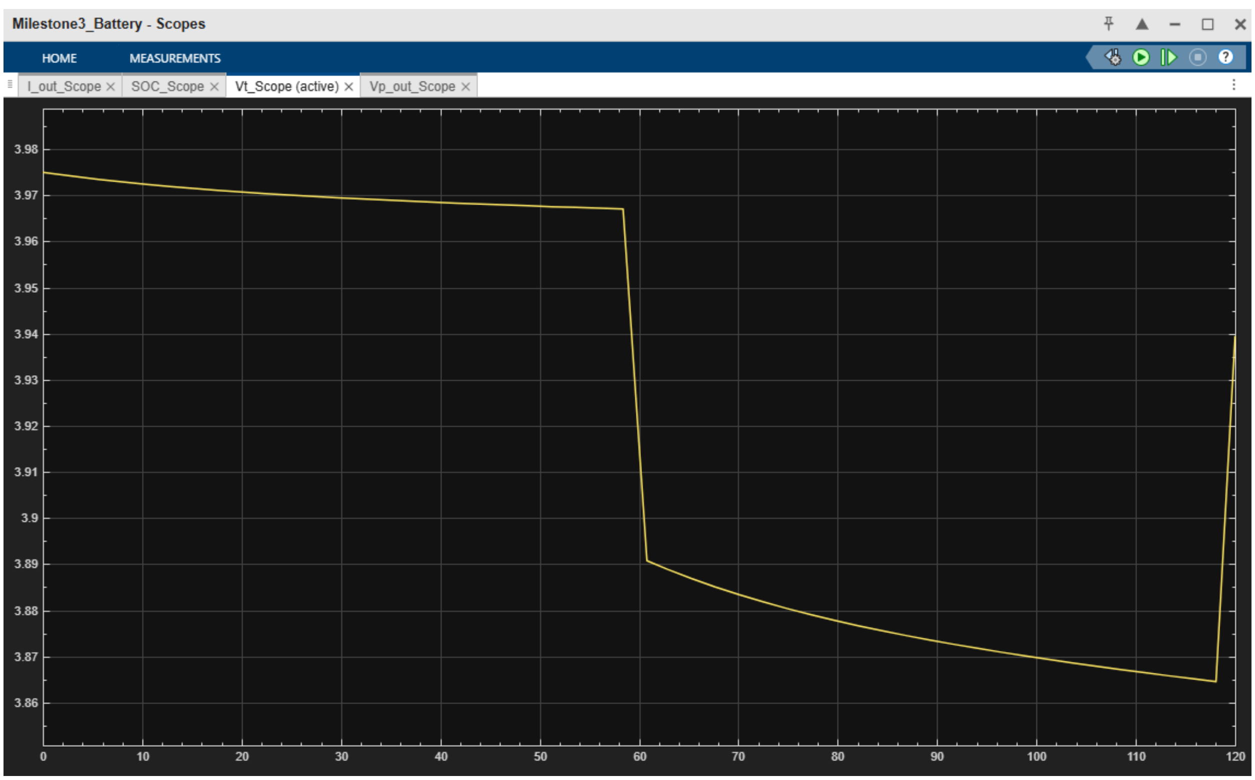
The I_profile in this case is equal to the output current since it does not change. The SOC gain was defined with SOC' = 1/Q_nom*3600, then it was integrated using a 1/s integrator to output the SOC. An Open-Circuit Voltage (OCV) lookup table was then defined with its dimensions being OCV_vec and SOC_vec, then it was added to the SOC output as the ONLY positive part of the Vt summation. The Vp had 2 gains, the first gain was Vp' = -1/R1*C1, and the second gain was Vp' = 1/C1. The 1/C1 gain was implemented directly on the input current and then integrated to output Vp, then the -1/R*C1 gain was implemented on the integrator

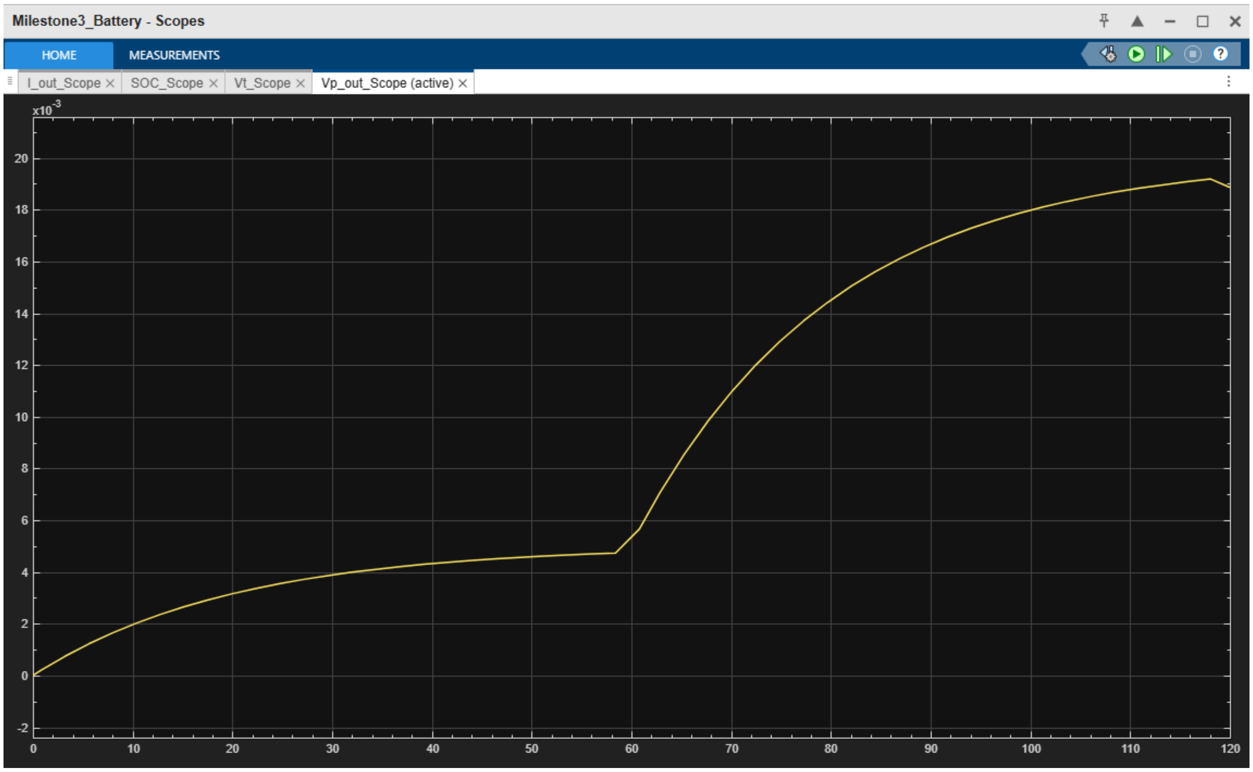
feedback and summated with the output of the 1/C1 gain. Finally, the resulting Vp_out voltage was outputted into a port and at the same time added as a one of the 2 negative parts of the Vt summation block. A resistance load was also added as the 2nd negative part of the Vt summation block in order to implement equation 3 of the model.

3. Graph outputs → The outputs of the model were scoped as the following graphs:




Milestone3_Battery - Scopes
HOME
MEASUREMENTS
I_out_Scope
SOC_Scope (active)
Vt_Scope
Vp_out_Scope
0.901
0.9
0.899
0.898
0.897
0.896
0.895
0.894
0.893
0.892
0.891
0 10 20 30 40 50 60 70 80 90 100 110 120


Milestone3_Battery - Scopes
HOME
MEASUREMENTS
I_out_Scope
SOC_Scope
Vt_Scope (active)
Vp_out_Scope
3.98
3.97
3.96
3.95
3.94
3.93
3.92
3.91
3.9
3.89
3.88
3.87
3.86
0 10 20 30 40 50 60 70 80 90 100 110 120



4.References:

[1] A. Author, B. Author, "A Review of SOC/SOH Estimation Methods for Li-Ion Batteries," Batteries, vol. 10, no. 1, 2023. Available: https://www.mdpi.com/2313-0105/10/1/34

[2] C. Researcher, D. Analyst, "Equivalent-Circuit Modelling of Battery Aging and Thermal Behaviour," Journal of Energy Storage, vol. 78, 2024. Available: https://www.sciencedirect.com/science/article/abs/pii/S2352152X24034947

[3] E. Scientist, F. Engineer, "Improved ECMs for Charge/Discharge Cycles Using DRT Analysis," Journal of Applied Electrochemistry, vol. 55, 2025. Available: https://link.springer.com/article/10.1007/s10800-025-02273-6